

CivicMitra – Acceptance Criteria & Edge Cases

1. Citizen Portal (User-Side)

■ *Acceptance Criteria*

- Citizens can submit reports with photo (optional), description, and auto GPS (or manual location).
- Each report generates a unique ticket ID.
- Users can track report statuses (Submitted → Acknowledged → In Progress → Resolved → Closed).
- Notifications (SMS/Push) are sent at each stage.
- App works as a PWA on Android, iOS, and desktop browsers.
- Supports multi-language (Hindi/English at MVP).
- Simple card-based navigation for categories.
- Offline support – reports stored locally and synced online.
- Citizens can reopen reports (within 48 hrs of closure).
- Duplicate issue warning if same location + category exists.

■■ *Edge Cases*

- No internet → report stored offline in queue.
- Photo missing → allowed, but report marked low context.
- GPS disabled → manual location entry enabled.
- Large images auto-compressed to <5MB.
- Spamming (50+ reports in 10 min) triggers warning/soft ban.
- Wrong issue type → backend auto-corrects classification.

2. Admin Portal (Role-Based Access)

Roles & Access

- SuperAdmin – manages platform, departments, officers, escalation rules.
- Department Admin – manages reports of their department, assigns tasks.
- Field Officer – executes tasks, updates statuses, uploads proof.

■ Acceptance Criteria – SuperAdmin

- Can add/remove/edit departments.
- Can onboard new officers/admins.
- Has full visibility of reports and analytics.
- Can manage escalation rules and SLAs.
- Can override/reassign tasks across departments.

■ Acceptance Criteria – Department Admin

- Views only department-specific reports.
- Can assign/reassign reports to officers.
- Updates status (Acknowledged → In Progress → Resolved).
- Can add remarks/notes visible to citizens.
- Generates department-specific analytics.

■ Acceptance Criteria – Field Officer

- Receives notifications for assigned tasks.
- Updates task progress (“On Site”, “Work Started”, “Work Completed”).
- Uploads photo proof of resolution.
- Can request reassignment for invalid reports.

■■ Edge Cases

- Auto-routing wrong → Admin can manually reassign.
- Officer unavailable → task escalates to backup officer.
- Simultaneous updates → latest update wins, conflicts logged.
- Unresolved reports beyond SLA → escalates to higher authority.
- Citizen reopens closed issue → reopened and reassigned.

3. Backend/System Layer

■ *Acceptance Criteria*

- All APIs secured with JWT authentication.
- Role-based access enforced across endpoints.
- System can handle 10k concurrent users with queues.
- Reports synced within 5 sec when online.
- Notifications triggered on status updates.
- Audit logs stored for all actions.
- PWA passes Lighthouse PWA audit (90+ score).
- Database encryption at rest + HTTPS enforced.

■■ *Edge Cases*

- Database crash → queued writes stored temporarily until recovery.
- Push notification failures retried with exponential backoff.
- Expired JWT → request denied, user re-authentication required.
- Simultaneous login from 2 devices → session throttling applied.
- Offline cache corruption → fallback to resync on next login.